BioTraj
User Manual
Version 1.2

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Contents

1 Introduction **3**

2 Software Architecture **3**

3 Installation **4**

4 Coordinate System **5**

5 Trajectory Extraction **5**
5.1 Overview . . . 5
5.2 Basic Usage . . . 5

6 Trajectory Data Format **5**

7 Trajectory Analysis **5**
7.1 Basic Usage . . . 5
7.2 Conventional Descriptors . . . 6
7.2.1 Explored Area and Monte Carlo Estimation . . . 6
7.2.2 Principal Component Analysis . . . 6
7.2.3 Mean-Squared Displacement . . . 7
7.2.4 Straightness and Fractal Dimension . . . 7

8 Information-Geometric Analysis **7**
8.1 Purpose and Interpretation . . . 7
8.2 Turning-Angle Distributions . . . 7
8.3 Fisher–Rao Distance . . . 8
8.4 Spatial Information Geometry . . . 8
8.5 Spatial Sampling Diagnostics . . . 9
8.6 Temporal Information Geometry . . . 10
8.7 Bootstrap and Finite-Sampling Controls . . . 11

9 Information-Geometry Command-Line Options **12**

10 Reproducing the Information-Geometry Settings Used in the BioTraj Study **13**
10.1 Synthetic Validation Data . . . 13

11 Information-Geometry Output Files **14**

12 Scene Annotation and Annotation-Aware Analysis **15**
12.1 Scene Annotation 15
12.2 Annotation-Aware Fisher–Rao Analysis 15
12.3 Annotation-Aware Workflow 16

13 Recommended Analysis Workflow **16**

14 Limitations **17**

15 Applications **17**

16 Citation and Reproducibility **17**

17 License **17**

1 Introduction

BioTraj (BioTrajectoryAnalysis) is an open-source collection of Python tools for extracting and quantitatively analysing biological trajectories from video recordings. The framework combines computer vision, automated tracking, conventional trajectory descriptors, and information-geometric analysis of spatially and temporally resolved movement distributions.

The software was initially developed for ant trajectory analysis, but the workflow can be adapted to other organisms and moving agents provided that suitable two-dimensional trajectories can be obtained. Tracking and analysis are separated through CSV trajectory files, so trajectories generated by other tracking programs can also be analysed.

The current release contains two principal components:

1. `AntTracking.py`: video-based trajectory extraction;
2. `AntTrajectoryAnalysis.py`: conventional trajectory analysis together with spatial and temporal Fisher–Rao information geometry.

Companion utilities provide scene annotation and annotation-aware analysis, while a synthetic trajectory generator provides controlled validation datasets. These utilities are described later in this manual.

2 Software Architecture

The BioTraj workflow is modular (Figure 1). Video acquisition and computer-vision tracking are separated from quantitative trajectory analysis through an intermediate CSV trajectory file. The tracking stage is implemented by `AntTracking.py`, whereas the principal analysis stage is implemented by `AntTrajectoryAnalysis.py`. This separation allows BioTraj to analyse trajectories produced by other tracking programs, provided that they are converted to the expected CSV representation, and permits new analysis methods to be added without modifying the tracking module.

The analysis engine provides conventional spatial and dynamical descriptors and, when requested, spatial and temporal information-geometric analyses. Scene annotation and annotation-aware Fisher–Rao analysis are implemented as companion modules described later in this manual.

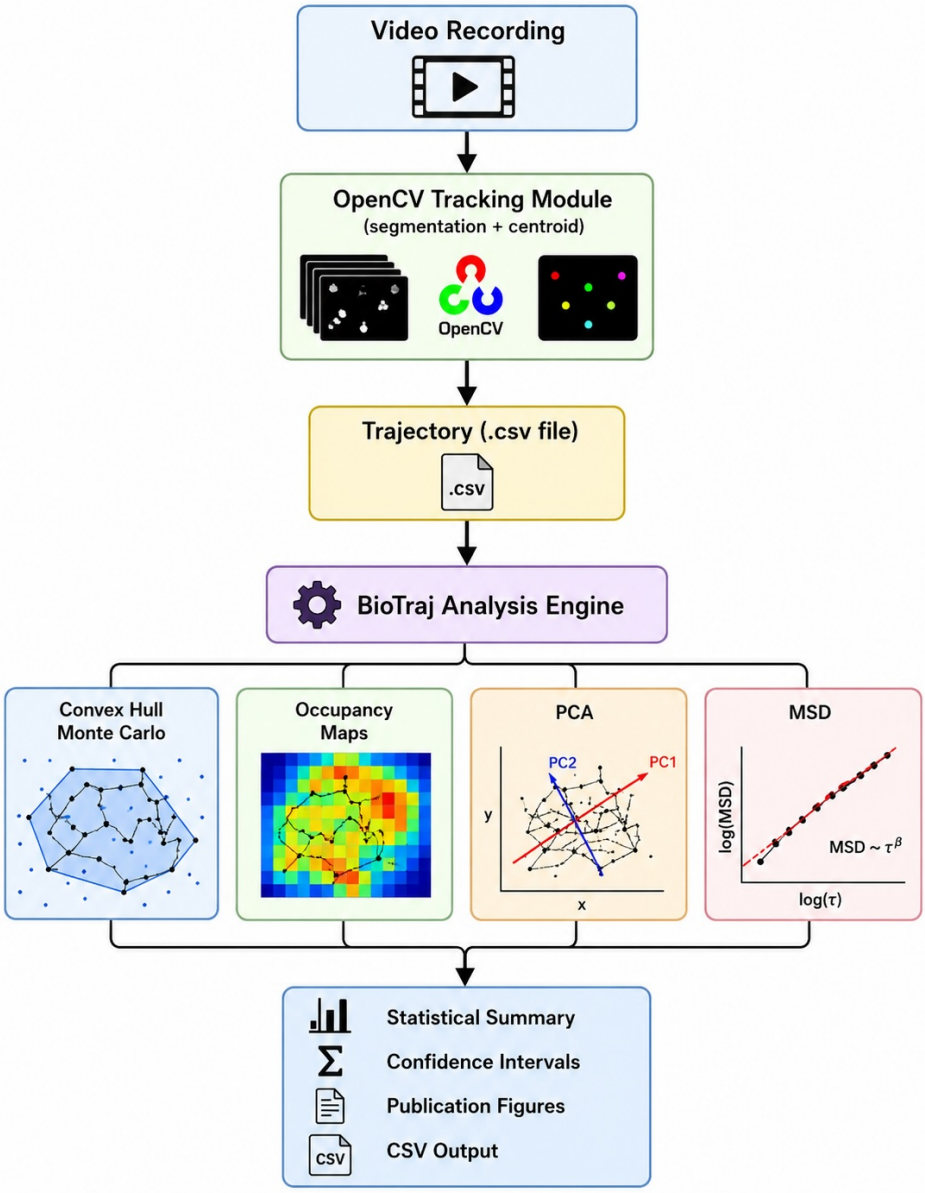


Figure 1: **BioTraj software architecture.** A video recording is processed by the computer-vision tracking module (`AntTracking.py`) to produce trajectory coordinates in CSV format. The trajectory file is then analysed by the BioTraj analysis engine (`AntTrajectoryAnalysis.py`). The diagram illustrates representative conventional outputs, including explored-area estimates, occupancy maps, PCA-based anisotropy and mean-squared-displacement analysis. Spatial and temporal Fisher–Rao information geometry and the companion annotation-aware analysis extend this core workflow and are described in later sections.

3 Installation

BioTraj requires Python 3.10 or newer. Principal dependencies include OpenCV, NumPy, Pandas, SciPy, Matplotlib, and scikit-learn. Install repository requirements using:

```
pip install -r requirements.txt
```

4 Coordinate System

Video images follow the OpenCV convention, with the origin at the upper-left corner. Statistical analyses and graphical outputs may transform coordinates to a Cartesian representation with the origin at the lower-left, the x axis increasing to the right, and the y axis increasing upwards. Users should retain the original image coordinate system when defining scene annotations intended to overlay the video frame.

5 Trajectory Extraction

5.1 Overview

The tracking program detects moving objects, filters detections, associates detections between frames, reconstructs continuous track segments, and writes coordinates to a CSV file. A tracking identifier denotes a continuous reconstructed track segment; it should not automatically be interpreted as a persistent biological individual when identities can be lost or exchanged.

5.2 Basic Usage

```
python3 AntTracking.py video.mov
```

A typical configured run is:

```
python3 AntTracking.py video.mov \
    --threshold 18 \
    --min-area 6 \
    --max-area 250 \
    --output tracking_results
```

Important tracking parameters include the detection threshold, minimum and maximum object area, maximum association distance, maximum permitted track gap, and merge distance. Appropriate values depend on image scale, contrast, frame rate, and organism size.

6 Trajectory Data Format

The analysis program accepts trajectory data in CSV format. A typical file contains:

```
track_id,frame,x,y
1,0,123.4,201.7
1,1,124.2,202.3
1,2,125.0,203.1
```

Coordinates may be expressed in pixels or calibrated physical units. Turning angles are calculated from successive stored positions after observations have been grouped by track identifier and ordered by frame number. The current implementation does not additionally require those observations to occur in consecutive video frames; therefore, if the same track identifier is retained across a short detection gap, a turning angle may span that gap.

7 Trajectory Analysis

7.1 Basic Usage

The current analysis program is run with the trajectory CSV as the positional argument:

```
python3 AntTrajectoryAnalysis.py \
    ant_tracks.csv \
    --output analysis_results
```

Without information-geometry switches, the program provides the conventional BioTraj trajectory descriptors and associated figures.

7.2 Conventional Descriptors

BioTraj calculates complementary geometric and dynamical descriptors, including convex-hull explored area, Monte Carlo area estimation, occupancy maps, principal-component anisotropy, mean-squared displacement (MSD), straightness, and fractal dimension. Figure 2 illustrates how these descriptors fit into the experimental workflow, from video recording and object tracking to reconstructed trajectories and quantitative analysis.

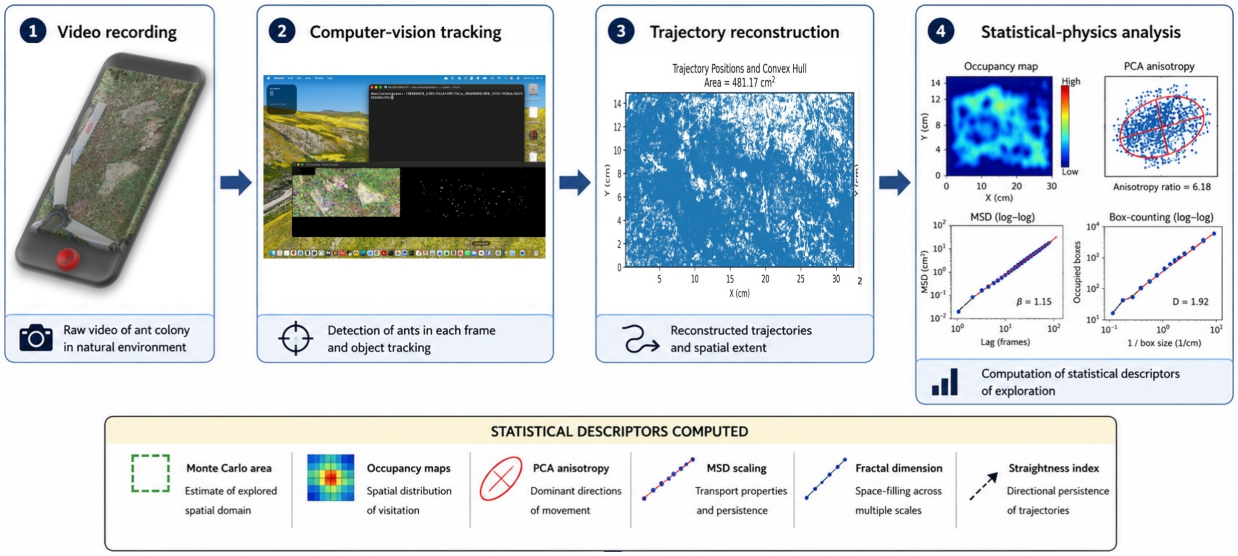


Figure 2: **From video recording to conventional BioTraj descriptors.** A biological recording is processed by computer-vision tracking to reconstruct trajectories and their spatial extent. BioTraj then calculates complementary descriptors of spatial exploration and movement dynamics. Representative outputs shown here include occupancy maps, PCA anisotropy, MSD scaling, box-counting fractal dimension, Monte Carlo area estimation and trajectory straightness. The figure illustrates the conventional-analysis branch of BioTraj; the information-geometric analyses described later provide additional spatially and temporally resolved comparisons of movement distributions.

7.2.1 Explored Area and Monte Carlo Estimation

The explored domain is approximated by the convex hull. A Monte Carlo estimate can be obtained from random points within a bounding rectangle:

$$A_{\mathrm{MC}} = A_{\mathrm{box}} \frac{N_{\mathrm{inside}}}{N_{\mathrm{total}}}. \tag{1}$$

7.2.2 Principal Component Analysis

For trajectory positions $\mathbf{r}_i$, the covariance matrix is diagonalized to obtain eigenvalues λ_1 and λ_2. The anisotropy ratio is

$$A = \frac{\lambda_1}{\lambda_2}. \tag{2}$$

Values near unity indicate approximately isotropic spatial exploration; larger values indicate an increasingly elongated spatial distribution.

7.2.3 Mean-Squared Displacement

$$\mathrm{MSD}(\tau) = \left\langle |\mathbf{r}(t+\tau) - \mathbf{r}(t)|^2 \right\rangle . \tag{3}$$

When an approximate power law $\mathrm{MSD}(\tau) \propto \tau^{\beta}$ is observed, $\beta < 1$ corresponds to sublinear/-subdiffusive scaling, $\beta \approx 1$ to diffusive scaling, and $\beta > 1$ to superlinear/superdiffusive scaling. Superlinear scaling can arise from persistent or otherwise correlated motion and should not be assigned a unique mechanism from β alone.

7.2.4 Straightness and Fractal Dimension

Straightness compares net displacement with total path length. Fractal dimension provides a complementary measure of trajectory complexity and space-filling character over the analysed scales.

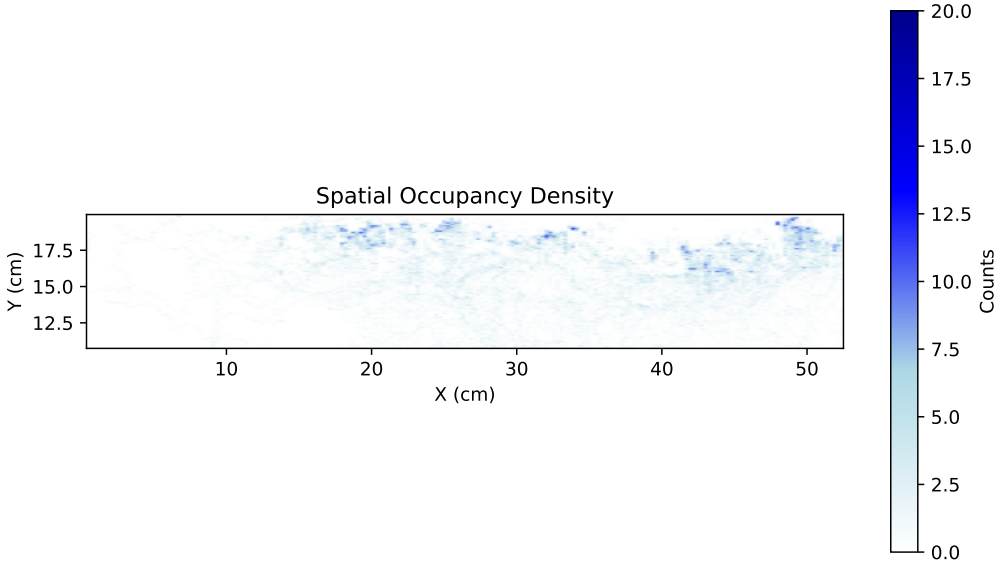


Figure 3: Example BioTraj occupancy map. Occupancy describes *where* observations occur and is complementary to the information-geometric analysis, which describes how local movement distributions differ.

8 Information-Geometric Analysis

8.1 Purpose and Interpretation

The information-geometric module detects spatial or temporal changes in empirical movement distributions without requiring predefined behavioural states. The present implementation uses unsigned turning-angle distributions.

Important interpretation rule: Fisher–Rao heterogeneity identifies where or when estimated movement statistics change; it does not determine why they change. Elevated values may reflect biological responses, physical constraints, surface geometry, projection effects, tracking artefacts, finite sampling, or combinations of these factors.

8.2 Turning-Angle Distributions

For three successive stored positions belonging to the same track identifier,

$$\Delta\mathbf{r}_t = \mathbf{r}_{t+1} - \mathbf{r}_t, \tag{4}$$

and, provided that both displacement vectors have non-zero length, the unsigned turning angle is

$$\Delta\theta_t = \arccos\left(\frac{\Delta\mathbf{r}_{t-1}\cdot\Delta\mathbf{r}_t}{\|\Delta\mathbf{r}_{t-1}\|\,\|\Delta\mathbf{r}_t\|}\right), \qquad 0 \le \Delta\theta \le \pi. \tag{5}$$

If either displacement vector has zero length, the angle is undefined and that observation is excluded. The angle is associated spatially with the middle position of the three-position sequence. Because the present representation is unsigned, it measures turning magnitude but does not retain left–right turning handedness.

Within a spatial cell or temporal window, the angles are converted to a normalized histogram on a fixed common angular partition,

$$P = (p_1, \ldots, p_K), \qquad p_k \ge 0, \qquad \sum_{k=1}^{K} p_k = 1. \tag{6}$$

8.3 Fisher–Rao Distance

The square-root embedding

$$\psi(P) = 2\,(\sqrt{p_1}, \ldots, \sqrt{p_K}) \tag{7}$$

maps the categorical probability simplex isometrically, under the Fisher metric convention used by BioTraj, to the positive orthant of a sphere of radius 2 because $\|\psi(P)\|_2 = 2$. Equivalently, the unscaled square-root vector $(\sqrt{p_1}, \ldots, \sqrt{p_K})$ lies on the unit sphere. Under this convention, the Fisher–Rao geodesic distance between two categorical distributions is

$$d_{\mathrm{FR}}(P, Q) = 2\arccos\left(\sum_{k=1}^{K}\sqrt{p_k q_k}\right). \tag{8}$$

Thus $0 \le d_{\mathrm{FR}} \le \pi$, with zero corresponding to identical distributions and π to distributions with disjoint support. The same fixed histogram partition must be used when distributions are compared.

8.4 Spatial Information Geometry

With `-info-geometry`, the analysed domain is divided into a rectangular grid. A turning-angle distribution is estimated for each sufficiently sampled cell. Pairwise Fisher–Rao distances quantify statistical separation between cells, while the local Fisher–Rao heterogeneity assigned to a cell summarizes its distances from valid neighbouring cells.

Large local values indicate that a cell's estimated turning-angle distribution differs strongly from those of its neighbours. Small values indicate locally similar distributions.

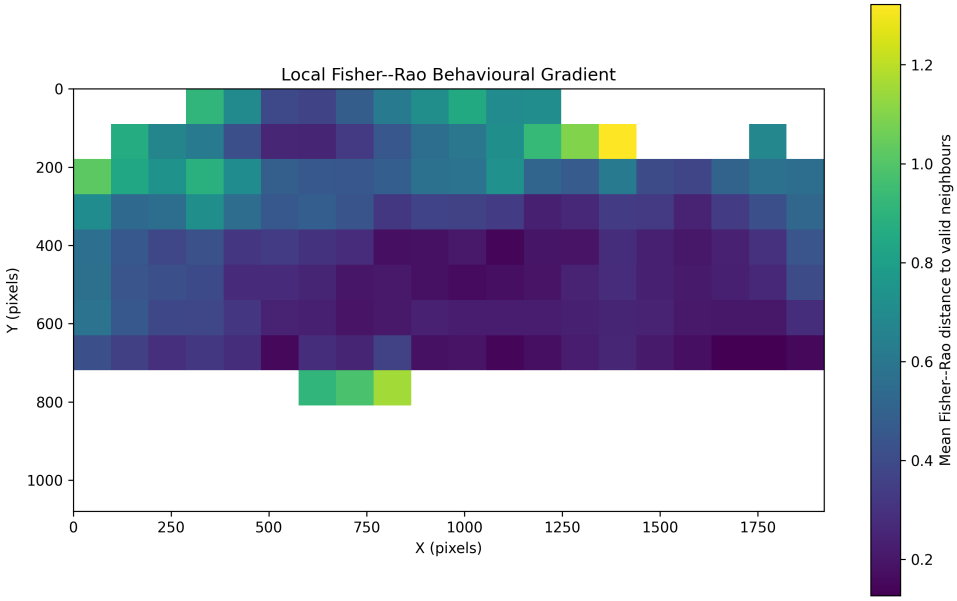


Figure 4: Example spatial Fisher–Rao map. Colour represents local Fisher–Rao heterogeneity calculated from neighbouring spatial cells. The map identifies spatial changes in movement statistics and should not be interpreted automatically as a behavioural-state map.

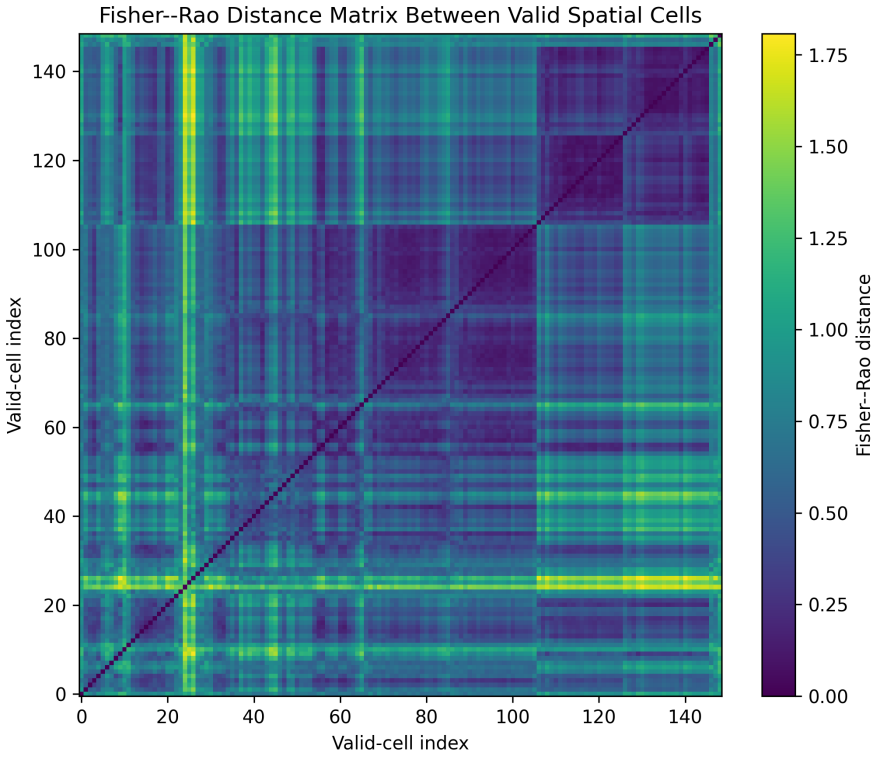


Figure 5: Example pairwise Fisher–Rao distance matrix between sufficiently sampled spatial cells. The matrix represents statistical similarity independently of physical separation; the local spatial map in Figure 4 restores neighbourhood context.

8.5 Spatial Sampling Diagnostics

Cells containing fewer than `-info-min-samples` valid turning angles are excluded. The sampling map should always be inspected together with the Fisher–Rao map because empirical histogram

distances become more variable when estimated from few observations.

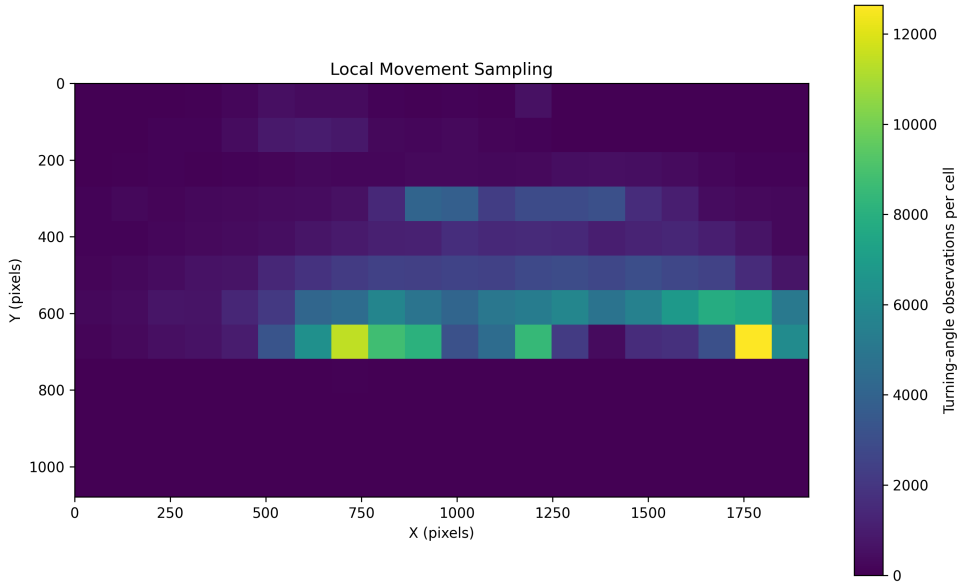


Figure 6: Example spatial sampling map showing the number of valid turning-angle observations available in each grid cell. Cells below the configured minimum sample count are excluded from information-geometric comparisons.

8.6 Temporal Information Geometry

With `-temporal-info-geometry`, the recording is divided into temporal windows. A turning-angle distribution is estimated for each sufficiently sampled window. BioTraj writes the complete temporal Fisher–Rao distance matrix and the distances between consecutive windows.

The window length controls temporal resolution. A sharp transition occurring inside a window produces a mixed distribution and may therefore yield the largest consecutive-window distance in the following comparison rather than exactly at the physical transition time.

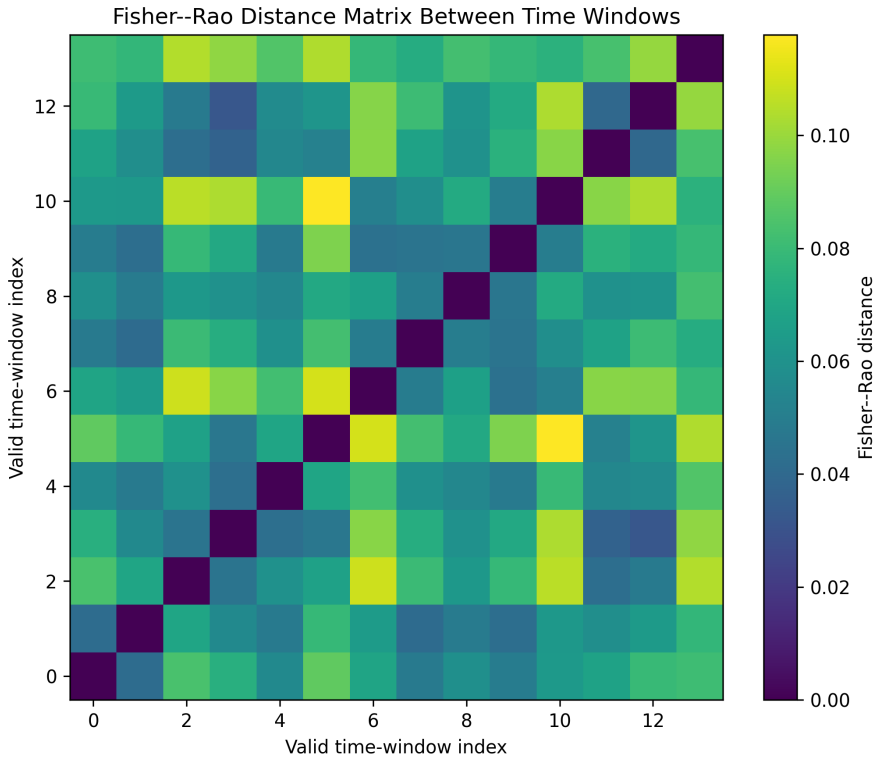


Figure 7: Example temporal Fisher–Rao distance matrix. Each element compares the turning-angle distributions estimated in two temporal windows.

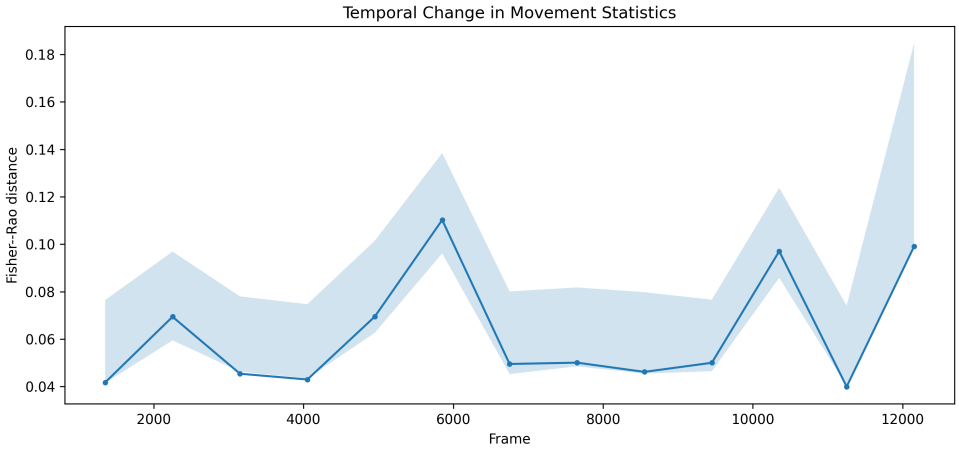


Figure 8: Example consecutive-window Fisher–Rao analysis. Peaks identify periods in which the estimated movement distribution changes strongly between adjacent temporal windows.

8.7 Bootstrap and Finite-Sampling Controls

The spatial and temporal modules can bootstrap empirical turning-angle samples to estimate uncertainty in Fisher–Rao distances. This quantifies empirical sampling variability but does not make temporally correlated observations or neighbouring spatial cells independent biological replicates.

Raw Fisher–Rao values should not be compared across datasets with very different local sample sizes without additional sampling controls. BioTraj therefore supports interpretation using complementary diagnostics. Equal-count resampling tests whether differences persist when the same number of observations is used in each eligible spatial cell, whereas threshold-convergence analysis examines the stability of the result as the minimum number of observations required for a cell is

increased. Figure 9 illustrates the equal-count approach.

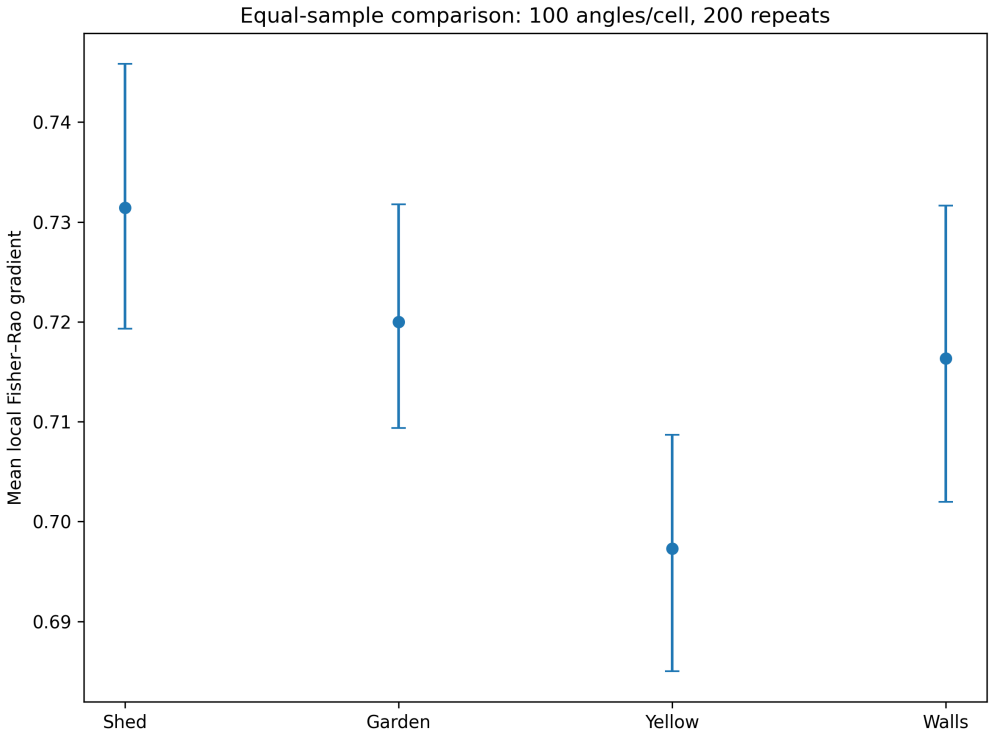


Figure 9: Equal-sample control for local Fisher–Rao heterogeneity. Turning-angle observations are resampled to a common sample size before calculating local Fisher–Rao quantities, reducing differences caused solely by unequal numbers of observations among spatial cells. Repeated resampling provides a measure of finite-sampling variability and helps distinguish sampling effects from persistent differences among recordings.

9 Information-Geometry Command-Line Options

The principal options in BioTraj v1.2 are summarized below.

Option	Default	Meaning
`-info-geometry`	off	Enable spatial information geometry.
`-info-grid-x`	20	Number of spatial grid columns.
`-info-grid-y`	12	Number of spatial grid rows.
`-info-angle-bins`	18	Number of bins over 0–180°.
`-info-min-samples`	30	Minimum turning-angle observations per spatial cell.
`-info-clusters`	4	Number of hierarchical information-geometric distributional clusters. Clustering is optional/descriptive and does not define biological states.
`-info-pseudocount`	0	Optional pseudocount added per angular bin.
`-info-bootstrap`	200	Spatial bootstrap replicates; 0 disables bootstrap.
`-info-seed`	12345	Random seed for information-geometry bootstrap.

Option	Default	Meaning
`-temporal-info-geometry`	off	Enable temporal information geometry.
`-temporal-window-frames`	1800	Temporal window length in frames.
`-temporal-step-frames`	window length	Step between temporal windows.
`-temporal-min-samples`	100	Minimum turning-angle observations per temporal window.
`-temporal-bootstrap`	200	Temporal bootstrap replicates; 0 disables bootstrap.
`-fps`	none	Frame rate; when supplied, temporal axes can be expressed in seconds.

10 Reproducing the Information-Geometry Settings Used in the BioTraj Study

The analyses reported with BioTraj v1.2 used a 20×12 spatial grid, 18 turning-angle bins, a minimum of 30 angles per spatial cell, 200 bootstrap replicates, and 15-s non-overlapping temporal windows for 60-fps recordings. Because the program default temporal window is 1800 frames, the 900-frame manuscript setting must be supplied explicitly:

```
python3 AntTrajectoryAnalysis.py \
    ant_tracks.csv \
    --output BioTraj_V1_2_results \
    --info-geometry \
    --info-grid-x 20 \
    --info-grid-y 12 \
    --info-angle-bins 18 \
    --info-min-samples 30 \
    --info-clusters 4 \
    --info-pseudocount 0 \
    --info-bootstrap 200 \
    --info-seed 12345 \
    --temporal-info-geometry \
    --temporal-window-frames 900 \
    --temporal-step-frames 900 \
    --temporal-min-samples 100 \
    --temporal-bootstrap 200 \
    --fps 60
```

10.1 Synthetic Validation Data

BioTraj v1.2 includes the controlled trajectory generator `BioTraj_SyntheticTrajectoryGenerator_v1_2_stru`. The generator produces synthetic trajectories for which the underlying spatial and/or temporal statistical structure is known in advance. These datasets provide a controlled test of the information-geometric analysis independently of experimental tracking uncertainty.

The generator contains five scenarios:

1. **Homogeneous stationary**: the same movement law is used throughout the arena and throughout the recording;
2. **Spatial two-domain**: the movement law changes across a known vertical spatial boundary;
3. **Temporal two-state**: the movement law changes at a known frame;
4. **Spatiotemporal**: spatial domains are combined with a known temporal change;

5. **Structured arena**: a localization benchmark containing a circular nest/entrance zone, a directional corridor, a wall/surface-transition boundary, and ordinary observation-frame exits.

The first four scenarios correspond to the controlled synthetic validation cases used in the BioTraj study. The structured-arena case is an additional validation example distributed with BioTraj v1.2. Its semantic geometry is stored as ground truth but is not supplied to the BioTraj information-geometric analysis.

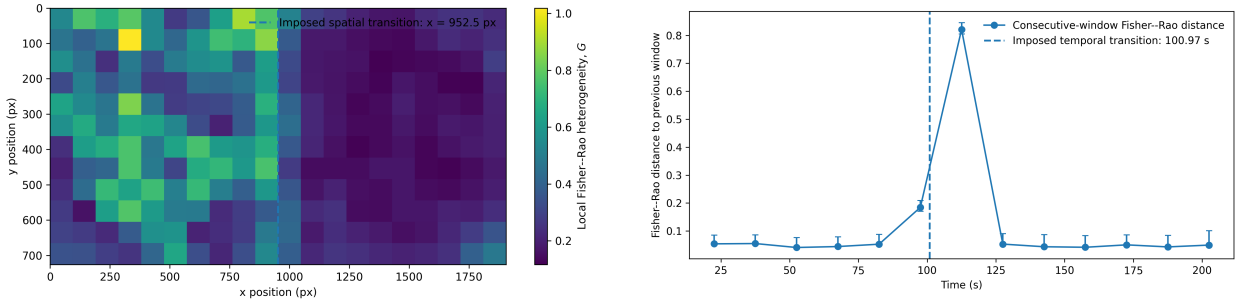
The synthetic movement model updates the direction according to a von Mises turning distribution and draws positive step lengths from a truncated normal distribution. It is intentionally a controlled validation model rather than a realistic mechanistic model of ant behaviour.

To generate all supplied validation scenarios without automatically launching the BioTraj analysis, use for example:

```
python3 BioTraj_SyntheticTrajectoryGenerator_v1_2.py \
    --all \
    --base-dir synthetic_validation \
    --seed 12345 \
    --generate-only
```

Each generated scenario contains an `ant_tracks.csv` file compatible with `AntTrajectoryAnalysis.py`, together with configuration and ground-truth metadata and a trajectory preview. The generated trajectories can therefore be passed through the same analysis workflow used for experimental data.

For the validation cases reported in the BioTraj study, the synthetic trajectories used 20 agents, 12,116 frames, a 1905×726 pixel arena, 60 frames s^{-1}, and random seed 12345. The imposed spatial and temporal changes are known independently of the information-geometric analysis, allowing recovery of these changes to be assessed at the resolution of the selected spatial grid and temporal windows.



(a) Spatial validation **(b) Temporal validation**

Figure 10: **Simulation-based validation of the information-geometric analysis.** (a) Spatial validation using synthetic trajectories with a known change in movement statistics across the arena. The spatial Fisher–Rao analysis is evaluated against the imposed spatial structure. (b) Temporal validation using synthetic trajectories with a known change in movement statistics during the recording. The temporal Fisher–Rao analysis is evaluated against the imposed transition time. Because the spatial and temporal changes are specified independently when the trajectories are generated, these tests provide controlled validation of the ability of BioTraj to recover distributional heterogeneity at the resolution of the selected spatial grid and temporal windows.

11 Information-Geometry Output Files

Depending on enabled options, the v1.2 analysis writes numerical tables, summaries, and figures. Important information-geometric outputs include:

- `info_geometry_cells.csv`: cell-level spatial information-geometry results;
- `info_geometry_summary.json`: spatial analysis settings and summary quantities;
- spatial sampling, local Fisher–Rao gradient, distance-matrix, and distributional-cluster figures;
- `temporal_info_windows.csv`: window-level temporal information;
- `temporal_fisher_rao_distance_matrix.csv`: complete temporal distance matrix;
- `temporal_info_geometry_summary.json`: temporal settings and summary information;
- temporal sampling, distance-matrix, and consecutive-distance figures.

File names can evolve slightly between development builds; the release version and its generated summary files should be retained with published analyses.

12 Scene Annotation and Annotation-Aware Analysis

The supplied `examples/WallAnts/` dataset illustrates the complete three-stage workflow. Tracking outputs are stored in `tracking_results/`, conventional and information-geometric analysis outputs in `BioTraj_V1_2_revision2/`, and the annotation-aware Fisher–Rao results in `WALL_annotated_FR_v1_1/`. The example also includes `FirstFrame.jpg`, the corresponding scene-annotation JSON file, and an annotated reference image.

BioTraj v1.2 includes two companion utilities for relating independently defined features of the recorded scene to the spatial Fisher–Rao results: `BioTraj_Scene_Annotator.py` and `BioTraj_Annotated_FR_An` These utilities are separate from the main trajectory-analysis program so that physical or biological scene information can be defined independently of the information-geometric result.

12.1 Scene Annotation

The graphical scene annotator can be started with:

```
python3 BioTraj_Scene_Annotator.py FirstFrame.jpg
```

where `FirstFrame.jpg` is a reference image in the same coordinate system as the tracked trajectories.

The annotator stores scene information in JSON format using original-image pixel coordinates. It can be used to define polygonal regions, polylines, circles, surface regions, surface transitions, biological features, physical constraints, observation boundaries, and artifact or exclusion regions.

Three concepts should remain distinct:

$$\textbf{tracking domain} \neq \textbf{biological/physical regions} \neq \textbf{artifact masks}.$$

Whenever possible, scene annotations intended for biological or physical interpretation should be defined independently of the Fisher–Rao map. This reduces the risk of selecting regions retrospectively because they coincide with features already visible in the information-geometric result.

12.2 Annotation-Aware Fisher–Rao Analysis

The annotation-aware analysis combines the scene-annotation JSON file with `info_geometry_cells.csv` generated by `AntTrajectoryAnalysis.py`. A typical command is:

```
python3 BioTraj_Annotated_FR_Analysis_v1_1.py \
    --annotations FirstFrame_scene_annotations.json \
    --info-cells BioTraj_results/info_geometry_cells.csv \
    --tracks ant_tracks.csv \
```

```
--image FirstFrame.jpg \
--grid-x 20 \
--grid-y 12 \
--output annotated_FR_results
```

The `-tracks` argument is optional but is recommended when the original BioTraj image frame should be inferred consistently. The `-image` argument is optional and supplies a reference image for registration and Fisher–Rao overlays. The explicit `-grid-x` and `-grid-y` values are strongly recommended when `info_geometry_cells.csv` contains only the populated/valid cells, because they preserve the full spatial grid used by the original analysis.

The program assigns valid information-geometric cells to annotated scene features and produces regional and transition-based summaries. It can also evaluate local Fisher–Rao heterogeneity as a function of distance from annotated transitions and generate graphical overlays and quality-control outputs. Regions marked for exclusion in the annotation file can be treated as artifact masks in the corresponding quality-controlled summaries.

Annotation-aware comparisons remain descriptive unless the experimental design provides genuine independent biological replication. Spatial cells within one recording should not automatically be treated as independent biological replicates, and a relationship between Fisher–Rao heterogeneity and an annotated feature does not by itself establish a biological mechanism.

12.3 Annotation-Aware Workflow

The intended sequence is:

1. extract trajectories with `AntTracking.py`, or provide compatible trajectory data from another tracker;
2. analyse the trajectories with `AntTrajectoryAnalysis.py` and generate `info_geometry_cells.csv`;
3. select a reference image, normally the first video frame;
4. define scene features independently using `BioTraj_Scene_Annotator.py`;
5. combine the annotation JSON file and the spatial information-geometric results using `BioTraj_Annotated_`
6. inspect both raw and quality-controlled regional/transition summaries before biological interpretation.

13 Recommended Analysis Workflow

A robust information-geometric workflow is:

1. inspect tracking quality and preserve continuous track segments;
2. inspect occupancy and conventional trajectory descriptors;
3. inspect the spatial sampling map;
4. calculate spatial and/or temporal Fisher–Rao results using justified minimum sample counts;
5. inspect bootstrap uncertainty and finite-sampling diagnostics;
6. relate detected statistical structure to independently defined scene information where appropriate;
7. avoid assigning biological mechanisms or behavioural states from Fisher–Rao values alone.

14 Limitations

The current implementation primarily assumes two-dimensional motion in the imaging plane. Tracking can be affected by poor contrast, object overlap, illumination changes, out-of-plane motion, and projection effects. Information-geometric estimates additionally depend on histogram resolution, spatial/temporal partitioning, and finite sample size. Spatial cells and temporal observations from a single recording are generally correlated and should not automatically be treated as independent biological replicates.

15 Applications

BioTraj can be adapted to ants and other social insects, protozoa, rotifers, nematodes, slugs and snails, small aquatic organisms, active-matter experiments, and simulated or robotic agents. Suitability depends on whether reliable trajectories can be reconstructed at a spatial and temporal resolution appropriate to the biological question.

16 Citation and Reproducibility

If BioTraj is used in academic work, cite the associated publication and the archived/released version of the BioTrajectoryAnalysis repository. For reproducibility, record the software release or commit, command-line options, random seed, input trajectory file, frame rate, calibration, and any scene-annotation files used in interpretation.

17 License

BioTraj is distributed under the MIT License.